

## Lecture 11

# SO(3), the double cover, and spherical harmonics

**Objectives.** Same algebra, second group. We prove that the adjoint map is a two-to-one surjection  $SU(2) \rightarrow SO(3)$  with kernel  $\{\pm\mathbb{I}\}$  — the double cover — and read off its topology:  $SO(3)$  is the three-sphere with antipodal points identified, and it is not simply connected. We determine which spin representations descend (exactly the integer ones), completing Lecture 9's ledger; we realise them concretely as harmonic polynomials and spherical harmonics, proving  $\dim = 2\ell + 1$ , irreducibility, the Laplace eigenvalue  $-\ell(\ell + 1)$  and completeness in  $L^2(S^2)$ ; we give Euler angles and Wigner matrices the single page they deserve; and we resolve Lecture 10's IOU: quantum mechanics acts on rays, Wigner's theorem makes that precise, and the physicist's rotation group is therefore  $SU(2)$ .

## 11.1 Two groups, one algebra

Lecture 8 computed the Lie algebra of the rotation group:  $\mathfrak{so}(3)$ , the antisymmetric  $3 \times 3$  matrices, with basis  $(L_a)_{jk} = -\epsilon_{ajk}$  and brackets  $[L_a, L_b] = \epsilon_{abc}L_c$  — the same structure constants as  $\mathfrak{su}(2)$ . The map

$$\psi: \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3), \quad \psi(X_a) = L_a,$$

is a Lie algebra isomorphism (Lecture 8), and Lecture 9 unmasked it:  $\psi = \text{ad}$ , the adjoint representation of  $\mathfrak{su}(2)$  in the orthonormal basis  $\{X_a\}$ . Two consequences follow at once. First, the representation theory of the algebra  $\mathfrak{so}(3)$  is already finished: transporting along  $\psi$  and complexifying (Lecture 10's lemma applies verbatim), the irreducible complex representations of  $\mathfrak{so}(3)$  are the spin- $j$  representations, one for every  $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ . Second, physics done infinitesimally cannot tell the two algebras apart: the Hermitian generators obey  $[J_a, J_b] = i\epsilon_{abc}J_c$  on either side of  $\psi$ . Whatever distinguishes  $SO(3)$  from  $SU(2)$  is invisible to commutation relations.

What distinguishes them is global, and Lecture 9 taught us its name. The ledger of integration (Table 9.1) has a blank row: is  $SO(3)$  simply connected, and which representations of its algebra integrate to the group? Today fills the row, and the instrument is the map  $\psi$  promoted from algebras to groups.

## 11.2 The double cover

**Theorem 11.1 (The double cover).** The adjoint map  $\Phi := \text{Ad}: SU(2) \rightarrow SO(3)$ , written in the orthonormal basis  $\{X_a\}$  as in Lecture 9, is a continuous surjective homomorphism with

$$\ker \Phi = \{\pm\mathbb{I}\}.$$

Consequently  $\Phi$  is exactly two-to-one —  $\Phi(U) = \Phi(V)$  if and only if  $V = \pm U$  — and Lecture 2's first isomorphism theorem gives

$$SO(3) \cong SU(2)/\{\pm \mathbb{I}\}. \quad (11.1)$$

*Proof.* That  $\Phi$  is a continuous homomorphism into  $SO(3)$  is Lecture 9. Surjectivity: Lecture 9 also gives  $\Phi(e^X) = e^{\text{ad}_X} = e^{\psi(X)}$  for  $X \in \mathfrak{su}(2)$ ; the isomorphism  $\psi$  is onto  $\mathfrak{so}(3)$ , and Lecture 8 proved the exponential map onto  $SO(3)$ , so every rotation is  $e^{\psi(X)} = \Phi(e^X)$ . Kernel:  $\Phi(U) = \mathbb{I}$  means  $UX_aU^{-1} = X_a$ , i.e.  $U$  commutes with each Pauli matrix. Commuting with  $\sigma_3$  forces  $U$  to preserve the eigenlines of  $\sigma_3$ , hence to be diagonal,  $U = \text{diag}(a, d)$ ; commuting with  $\sigma_1$ , which swaps the two basis vectors, forces  $a = d$ ; so  $U = a\mathbb{I}$  with  $a^2 = \det U = 1$ , giving  $U = \pm \mathbb{I}$ . Finally  $\Phi(U) = \Phi(V)$  iff  $V^{-1}U \in \ker \Phi$  iff  $U = \pm V$ . ■

**Remark 11.2 (A quotient that lands on its feet).** Equation (11.1) is the only place Part II touches an abstract quotient group. Quotients of matrix groups need not be matrix groups — nothing in Lecture 2's construction produces matrices — and our framework demands closed matrix groups. Here we are lucky twice over: the quotient is not merely isomorphic to some group, it is concretely realised, by  $\Phi$  itself, as the matrix group  $SO(3)$ . The abstract detour lasts one sentence.

Geometry first, then topology. Under the identification  $SU(2) \cong S^3$  of Lecture 10, the fibres of  $\Phi$  are the pairs  $\{\pm U\}$  — antipodal points of the sphere. So as a metric space,  $SO(3)$  is the three-sphere with antipodal points glued, each rotation owning exactly two antipodal addresses upstairs. The halved angles of Lectures 8–10 are this picture in coordinates (Figure 11.1): as  $\theta$  runs from 0 to  $2\pi$ , the rotation  $R_z(\theta)$  walks a closed loop downstairs while its address  $e^{-i\theta\sigma_3/2}$  walks only half of a great circle upstairs, from  $\mathbb{I}$  to  $-\mathbb{I}$ .

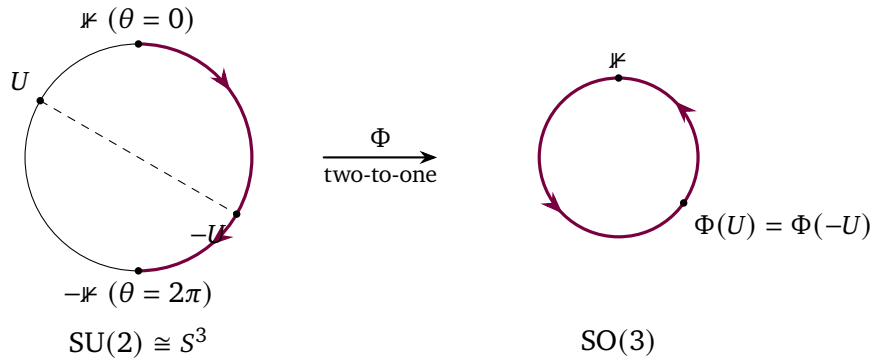


Figure 11.1: The double cover. Left: a great circle  $\theta \mapsto e^{-i\theta\sigma_3/2}$  of  $SU(2) \cong S^3$ ; as  $\theta$  runs from 0 to  $2\pi$  it traces the accented *open* path from  $\mathbb{I}$  to  $-\mathbb{I}$  — half the circle, the halved angle at work. Right: its image under  $\Phi$ , the *closed* loop  $R_z(\theta)$  in  $SO(3)$ , traversed once in full. Antipodal points  $\pm U$  (dashed chord) share a single image:  $\Phi$  glues opposite points of the sphere. The open lift is the obstruction of Proposition 11.3 — contracting the loop downstairs would have to drag the lift's endpoint through the discrete fibre  $\{\pm \mathbb{I}\}$  — while doubling the range to  $4\pi$  closes the lift, which then contracts in  $S^3$ .

**Proposition 11.3 (The loops of  $SO(3)$ ).**  $SO(3)$  is connected (Lecture 8) but not simply connected: the loop  $\rho(t) = R_z(2\pi t)$ ,  $t \in [0, 1]$ , is not contractible. Traversed twice, however, it

is. (That every loop in  $SO(3)$  shares this fate — the obstruction is exactly a sign — we state without proof.)

*Proof (sketch; the patching follows Lecture 9).* Doubling is the easy half. The loop  $\rho$  traversed twice is the  $\Phi$ -image of  $u(t) = e^{-2\pi it \sigma_3}$ , which is a genuine loop in  $SU(2)$  ( $u(0) = u(1) = \mathbb{1}$ ). Lecture 9 contracts  $u$  inside  $S^3$ ; composing that homotopy with the continuous  $\Phi$  contracts  $\rho^2$  inside  $SO(3)$ , basepoint fixed.

For the other half, lift. Around any point of  $SU(2)$  the map  $\Phi$  is locally invertible —  $\Phi(e^X) = e^{\psi(X)}$  together with Lecture 8's exponential charts shows it is a local homeomorphism with two continuous local inverses, differing by the sign of the sheet — so paths and homotopies in  $SO(3)$  lift to  $SU(2)$  by exactly the subdivide-and-patch argument that lifted circle loops in Lecture 9. The lift of  $\rho$  starting at  $\mathbb{1}$  is  $e^{-i\pi t \sigma_3}$ , ending at  $-\mathbb{1}$ : the loop opens up upstairs. If a homotopy contracted  $\rho$ , its lift would move the endpoint of the lifted path continuously through the discrete fibre  $\{\pm\mathbb{1}\}$  — hence not at all — while connecting it to the lift of the constant loop, whose endpoint is  $+\mathbb{1}$ . Contradiction. ■

**Remark 11.4 (The belt trick).** Proposition 11.3 can be held in the hands. Give a belt a full  $2\pi$  twist while keeping both ends fixed: no amount of manoeuvring the strap (= homotopy of the path of frames) removes the twist. Give it  $4\pi$ , and the twist can be undone — Dirac's belt trick (Figure 11.2). Notebook 11 animates both, next to the corresponding paths in  $SU(2)$ .

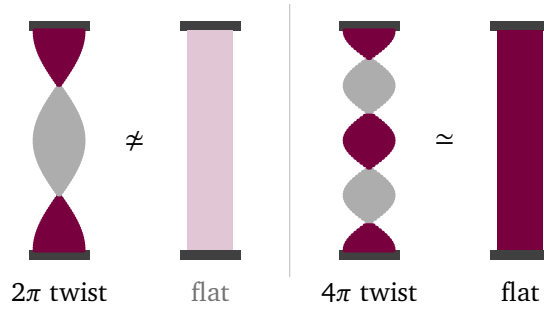


Figure 11.2: Dirac's belt trick, the topology of Proposition 11.3 held in the hands. A strap with both ends clamped records a path of frames by its twist; manoeuvring the strap without rotating the ends is a homotopy. A single  $2\pi$  twist (left) cannot be removed — it is not homotopic to the untwisted strap, just as  $\rho(t) = R_x(2\pi t)$  is not contractible in  $SO(3)$ . A  $4\pi$  twist (right) *can* be worked out flat by passing the strap over the clamped end, exactly as  $\rho^2$  contracts. The front of the strap is accented, its back grey, so each half-twist shows as a colour flip pinched to an edge.

So  $SU(2)$  and  $SO(3)$  share an algebra but not their loops, and by Lecture 9 that is precisely the difference that representation theory feels.

### 11.3 Which representations descend

**Theorem 11.5 (Descent through the kernel).** *Composition with  $\Phi$  is a bijection between continuous finite-dimensional representations of  $SO(3)$  and those representations  $\Pi$  of  $SU(2)$  with  $\Pi(-\mathbb{1}) = \mathbb{1}$ , matching invariant subspaces and irreducibility. In particular, since  $\Pi_j(-\mathbb{1}) =$*

$(-1)^{2j} \mathbb{K}$  (Lecture 10), the irreducible representations of  $SO(3)$  are exactly the integer-spin ones,

$$\Pi_\ell, \quad \ell = 0, 1, 2, \dots, \quad \dim \Pi_\ell = 2\ell + 1 = 1, 3, 5, \dots,$$

and no continuous representation of  $SO(3)$  contains half-integer spin.

*Proof.* If  $\tilde{\Pi}$  represents  $SO(3)$ , then  $\Pi = \tilde{\Pi} \circ \Phi$  is a continuous representation of  $SU(2)$  killing  $\ker \Phi$ ; since  $\Phi$  is onto, the two have the same invariant subspaces, and  $\Pi$  determines  $\tilde{\Pi}$ . Conversely, if  $\Pi(-\mathbb{K}) = \mathbb{K}$  then  $\Pi$  is constant on the fibres  $\{\pm U\}$ , so there is a unique  $\tilde{\Pi}$  with  $\tilde{\Pi} \circ \Phi = \Pi$ ; it is a homomorphism because  $\Phi$  is onto, and continuous because  $\Phi$  has continuous local inverses (the local sections of Proposition 11.3's proof), so that locally  $\tilde{\Pi} = \Pi \circ s$ . Lecture 10's list and sign  $(-1)^{2j}$  finish the classification. ■

Lecture 9's ledger closes:  $SO(3)$  is *not* simply connected, and the algebra representations that integrate to it are exactly those built from integer spins. The algebra could not see this —  $\psi$  is an isomorphism — the loop legislated. And the verdict agrees with analysis: Lecture 9 derived integer quantisation of orbital angular momentum from single-valuedness on a circle of rotations; today's theorem is the same fact read globally, since function actions are honest  $SO(3)$  representations and must therefore carry integer spin only. Where half-integers live we already know (Lecture 10); *why they are lawful* is Section 11.6. First, the integer representations deserve their concrete home.

## 11.4 Spherical harmonics: the integer representations, made of functions

Lecture 10 built every spin from polynomials in two complex variables. The integer spins admit a second construction, on the physicist's home ground: polynomials on  $\mathbb{R}^3$ , restricted to the sphere. Write  $(x, y, z)$  for coordinates on  $\mathbb{R}^3$ ,  $r = |x|$ , and let

$$P_\ell = \{\text{homogeneous complex polynomials of degree } \ell \text{ in } x, y, z\}, \quad \dim P_\ell = \frac{(\ell+1)(\ell+2)}{2},$$

(count the monomials  $x^a y^b z^c$  with  $a + b + c = \ell$ ). The function action  $(\Pi(R)f)(x) = f(R^{-1}x)$  of Lecture 3 preserves degree and acts continuously: a representation of  $SO(3)$  on  $P_\ell$ . One operator tells the spaces apart.

**Lemma 11.6 (The Laplacian is a rotation invariant).** For every  $R \in O(3)$  and smooth  $f$ ,  $\Delta(f \circ R^{-1}) = (\Delta f) \circ R^{-1}$ .

*Proof.* For a linear map  $A$  the chain rule gives  $\partial_a(f \circ A) = \sum_b A_{ba} (\partial_b f) \circ A$ , hence

$$\Delta(f \circ A) = \sum_{a,b,c} A_{ba} A_{ca} (\partial_b \partial_c f) \circ A = \sum_{b,c} (AA^T)_{bc} (\partial_b \partial_c f) \circ A,$$

and  $AA^T = \mathbb{K}$  for orthogonal  $A$ . ■

**Definition 11.7 (Harmonic polynomials).**  $H_\ell = \{f \in P_\ell : \Delta f = 0\}$ . By Lemma 11.6,  $H_\ell$  is an invariant subspace of  $P_\ell$ .

How big is it? The answer drops out of an inner product tailored to differentiation — the same trick as Lecture 10's Gaussian, in differential clothing. For  $f = \sum_\alpha c_\alpha x^\alpha$  write  $\tilde{f}(\partial) = \sum_\alpha \tilde{c}_\alpha \partial^\alpha$ , and on each  $P_\ell$  define

$$\langle f, g \rangle_\partial = (\tilde{f}(\partial) g)(0). \quad (11.2)$$

Monomials are orthogonal with  $\langle x^\alpha, x^\alpha \rangle_\partial = \alpha! > 0$ , so (11.2) is an inner product; and since multiplying a polynomial by  $r^2 = x^2 + y^2 + z^2$  corresponds, after the substitution  $x \mapsto \partial$ , to applying  $\Delta$ , it obeys the adjunction

$$\langle r^2 f, g \rangle_\partial = \langle f, \Delta g \rangle_\partial, \quad f \in P_{\ell-2}, \quad g \in P_\ell.$$

**Proposition 11.8 (Dimension of  $H_\ell$ ).**  $\Delta: P_\ell \rightarrow P_{\ell-2}$  is surjective, and

$$P_\ell = H_\ell \oplus r^2 P_{\ell-2} \quad (\text{orthogonal for (11.2)}), \quad \dim H_\ell = 2\ell + 1.$$

Iterating,  $P_\ell = H_\ell \oplus r^2 H_{\ell-2} \oplus r^4 H_{\ell-4} \oplus \cdots$ , so on the sphere ( $r = 1$ ) every polynomial restricts to a sum of restricted harmonics.

*Proof.* If  $h \in P_{\ell-2}$  is orthogonal to the range of  $\Delta$ , the adjunction gives  $\langle r^2 h, g \rangle_\partial = 0$  for all  $g \in P_\ell$ , so  $r^2 h = 0$  and hence  $h = 0$ :  $\Delta$  is surjective, and  $\dim H_\ell = \dim P_\ell - \dim P_{\ell-2} = \frac{(\ell+1)(\ell+2) - \ell(\ell-1)}{2} = 2\ell + 1$ . The adjunction also identifies  $\ker \Delta$  with the orthogonal complement of  $r^2 P_{\ell-2}$ , giving the decomposition; iterate on the second summand. ■

The dimension  $2\ell + 1$  is no coincidence, and the ladder machinery proves it. Assemble Lecture 9's generators for the three axes into Hermitian operators and ladders,

$$L_z = -i(x\partial_y - y\partial_x), \quad L_\pm = L_x \pm iL_y = \pm z(\partial_x \pm i\partial_y) \mp (x \pm iy)\partial_z,$$

which satisfy Lecture 10's ladder brackets  $[L_z, L_\pm] = \pm L_\pm$ ,  $[L_+, L_-] = 2L_z$  — same structure constants, same theorems.

**Theorem 11.9 ( $H_\ell$  is the spin- $\ell$  representation).**  $H_\ell$  is irreducible, of spin  $\ell$ , with highest-weight vector

$$w_\ell = (x + iy)^\ell.$$

*Proof.* First,  $w_\ell$  lives in  $H_\ell$  and is highest of weight  $\ell$ :

$$\Delta(x + iy)^\ell = \ell(\ell - 1)(x + iy)^{\ell-2} (1^2 + i^2) = 0, \quad L_z w_\ell = \ell w_\ell, \quad L_+ w_\ell = 0,$$

the last two because  $(\partial_x + i\partial_y)w_\ell = \ell(x + iy)^{\ell-1}(1 + i^2) = 0$  and  $\partial_z w_\ell = 0$ . Now run Lecture 10's descent inside  $H_\ell$  from  $w_\ell$ : the string formulas held in any representation (their derivation used only the brackets), so  $v_k = L_-^k w_\ell$  satisfies  $L_z v_k = (\ell - k)v_k$  and  $L_+ v_k = k(2\ell - k + 1)v_{k-1}$ ; the nonzero  $v_k$  are independent, and if  $v_{N+1} = 0 \neq v_N$  then  $0 = (N + 1)(2\ell - N)v_N$  forces  $N = 2\ell$  exactly. The span of  $v_0, \dots, v_{2\ell}$  is therefore a  $(2\ell + 1)$ -dimensional invariant subspace of  $H_\ell$ , irreducible by the simple-spectrum ladder argument of Lecture 10's polynomial model — and by Proposition 11.8 it is  $H_\ell$ . ■

**Definition 11.10 (Spherical harmonics).**  $Y_{\ell m}$  is the restriction to  $S^2$  of the basis vector of  $H_\ell$  of weight  $m$ , normalised in  $L^2(S^2)$  and phased by the ladder convention of Lecture 10. Concretely:  $Y_{\ell \ell} \propto (x + iy)^\ell|_{S^2}$  and  $Y_{\ell, m-1} \propto L_- Y_{\ell m}$ .

For  $\ell = 1$  the whole construction fits in one line:

$$w_1 = x + iy, \quad L_- w_1 = -2z, \quad L_-^2 w_1 = -2(x - iy), \quad L_-^3 w_1 = 0,$$

so  $Y_{1, \pm 1} \propto (x \pm iy)|_{S^2}$  and  $Y_{1,0} \propto z|_{S^2}$  — the  $p$ -orbitals of chemistry, generated rather than tabulated. Sheet 11 runs  $\ell = 2$  and compares with the standard tables. Note also, free of charge from homogeneity: under the parity map  $x \mapsto -x$  (which commutes with rotations),  $H_\ell$  has the definite sign  $(-1)^\ell$  — a fact Lecture 12's selection rules will spend.

So far the sphere has been a place to restrict polynomials. The restriction in fact carries all the structure: a homogeneous polynomial vanishing on  $S^2$  vanishes identically (homogeneity:  $u(x) = r^\ell u(x/r)$ ), so  $H_\ell$  maps bijectively onto

$$\mathcal{Y}_\ell := \{u|_{S^2} : u \in H_\ell\} \subset L^2(S^2),$$

and the bijection intertwines the rotation actions. Two theorems make  $\mathcal{Y}_\ell$  the centrepiece of mathematical physics: its elements diagonalise the sphere's Laplacian, and together they fill  $L^2(S^2)$ .

**Definition 11.11 (The spherical Laplacian).** For  $g$  smooth on  $S^2$ , let  $(Eg)(x) = g(x/|x|)$  be its degree-zero homogeneous extension, and define  $\Delta_{S^2} g := (\Delta Eg)|_{S^2}$ . (In spherical coordinates this is the familiar angular operator  $\sin^{-1}\theta \partial_\theta(\sin\theta \partial_\theta) + \sin^{-2}\theta \partial_\varphi^2$ ; the verification is Notebook 11's.)

**Theorem 11.12 (The eigenvalue  $-\ell(\ell+1)$ ).** Every  $Y \in \mathcal{Y}_\ell$  satisfies

$$\Delta_{S^2} Y = -\ell(\ell+1) Y. \quad (11.3)$$

*Proof.* Write  $Y = u|_{S^2}$  with  $u \in H_\ell$ ; homogeneity gives  $u = r^\ell EY$ . For  $h$  homogeneous of degree zero, Euler's identity  $(x \cdot \nabla)h = 0$ ; differentiate  $h(tx) = h(x)$  at  $t = 1$  kills the cross term in

$$\Delta(r^\ell h) = (\Delta r^\ell) h + 2 \nabla r^\ell \cdot \nabla h + r^\ell \Delta h = \ell(\ell+1) r^{\ell-2} h + r^\ell \Delta h,$$

using  $\partial_a r^\ell = \ell r^{\ell-2} x_a$ , hence  $\Delta r^\ell = \ell(\ell+1) r^{\ell-2}$ . Applying this to  $h = EY$  and  $u \in H_\ell$ :  $0 = \Delta u = \ell(\ell+1) r^{\ell-2} EY + r^\ell \Delta(EY)$ . Since  $\Delta(EY)$  is homogeneous of degree  $-2$ , restricting to  $r = 1$  gives (11.3). ■

**Remark 11.13 (Analysis meets algebra).** The eigenvalue is Lecture 10's Casimir at  $j = \ell$ . Indeed  $\mathcal{Y}_\ell$  is the spin- $\ell$  representation, on which  $L^2 = L_x^2 + L_y^2 + L_z^2$  acts as  $\ell(\ell+1) \mathbb{I}$  (Lecture 10's proposition, whose proof used only brackets), so on every  $\mathcal{Y}_\ell$  — and hence, by the completeness theorem below, on a dense subspace of  $L^2(S^2)$  —

$$-\Delta_{S^2} = L_x^2 + L_y^2 + L_z^2.$$

The spherical Laplacian is the Casimir of the rotation group; the Laplace operator has been doing representation theory since the eighteenth century.

**Theorem 11.14 (Spherical harmonics are complete).** The spaces  $\mathcal{Y}_\ell$  are mutually orthogonal in  $L^2(S^2)$ , and their closed span is everything:

$$L^2(S^2) = \widehat{\bigoplus_{\ell \geq 0} \mathcal{Y}_\ell}, \quad (11.4)$$

so the functions  $Y_{\ell m}$ ,  $\ell \geq 0$ ,  $-\ell \leq m \leq \ell$ , are an orthonormal basis of  $L^2(S^2)$ .

*Proof.* The rotation action on  $L^2(S^2)$  is unitary: the surface measure is rotation-invariant (substitution). Let  $\mathcal{P}_N$  be the space of restrictions to  $S^2$  of polynomials of degree  $\leq N$ : finite-dimensional, invariant, and — by Proposition 11.8 with  $r \equiv 1$  on the sphere — equal to  $\sum_{\ell \leq N} \mathcal{Y}_\ell$ . On  $\mathcal{P}_N$  we have an honest finite-dimensional unitary representation, so its generators are anti-Hermitian (Lecture 9), the Casimir  $L^2$  is Hermitian, and the  $\mathcal{Y}_\ell$  are its eigenspaces with the distinct eigenvalues  $\ell(\ell+1)$ : mutually orthogonal, with direct sum  $\mathcal{P}_N$ . Inside each  $\mathcal{Y}_\ell$  the ladder proposition of Lecture 10 makes the  $Y_{\ell m}$  orthonormal.

Density: the restricted polynomials form a subalgebra of  $C(S^2)$  containing the constants, separating points (the coordinates do) and closed under conjugation, hence uniformly dense by Stone–Weierstrass — imported here from the analysis course — and uniform density implies  $L^2$ -density on a finite measure space, with  $C(S^2)$  dense in  $L^2(S^2)$  from functional analysis. An orthonormal family with dense span is a basis. ■

The physics writes itself. A particle in a central potential,  $H = -\frac{1}{2\mu}\Delta + V(r)$ , commutes with rotations (Lemma 11.6); separating  $\psi = f(r)Y_{\ell m}$  and using Theorem 11.12 in the same computation that proved it,

$$-\frac{1}{2\mu}\left(f'' + \frac{2}{r}f'\right) + \left(V(r) + \frac{\ell(\ell+1)}{2\mu r^2}\right)f = E f :$$

the centrifugal barrier of every quantum mechanics course is a Casimir eigenvalue. Each radial solution arrives  $(2\ell + 1)$ -fold degenerate — Lecture 9’s degeneracy doctrine with the dimensions now computed — and *only* integer  $\ell$  occurs, by Theorem 11.5. Hydrogen’s spectrum is degenerate beyond this,  $n^2$  rather than  $2\ell + 1$ : a starred mystery (a hidden symmetry generated by the Runge–Lenz vector enlarges  $SO(3)$  to  $SO(4)$ ; optional project).

## 11.5 Euler angles and Wigner matrices

Computations with rotations need coordinates, and three angles suffice.

**Proposition 11.15 (Euler angles).** *Every  $R \in SO(3)$  can be written*

$$R = R_z(\alpha) R_y(\beta) R_z(\gamma), \quad \alpha, \gamma \in [0, 2\pi), \beta \in [0, \pi]. \quad (11.5)$$

*Proof.* The unit vector  $R\hat{z}$  has polar angles  $(\beta, \alpha)$  for some  $\beta \in [0, \pi]$ ,  $\alpha \in [0, 2\pi)$  (Figure 11.3), and  $R_z(\alpha)R_y(\beta)\hat{z}$  is exactly that vector. Hence  $Q = (R_z(\alpha)R_y(\beta))^{-1}R$  fixes  $\hat{z}$ , so it preserves the  $xy$ -plane, restricts there to an element of  $SO(2)$ , and equals  $R_z(\gamma)$  for some  $\gamma$ . (At  $\beta = 0$  or  $\pi$  only  $\alpha \pm \gamma$  is determined — the chart has a seam, known to every gimbal; Sheet 11 examines it.) ■

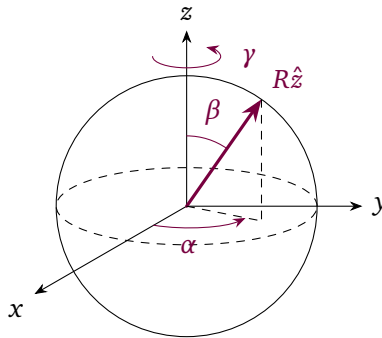


Figure 11.3: Euler angles. The pair  $(\beta, \alpha)$  are the polar coordinates of  $R\hat{z}$ , the image of the north pole —  $\beta$  measured down from the  $z$ -axis,  $\alpha$  around the equatorial plane from the  $x$ -axis (the dashed lines project  $R\hat{z}$  onto that plane). These two angles fix  $R_z(\alpha)R_y(\beta)$ , which carries  $\hat{z}$  to the right place; the leftover rotation about the original  $z$ -axis (upper circular arrow) is  $\gamma$ , the factor applied first in (11.5). At  $\beta = 0$  or  $\pi$  the azimuth degenerates — the gimbal’s seam.



**Definition 11.16 (Wigner matrices).**  $D_{m'm}^j(U) = \langle j, m' | \Pi_j(U) | j, m \rangle$ , the matrix elements of the irreducible representations in the ladder basis. Each  $D^j$  is a unitary  $(2j+1) \times (2j+1)$  matrix with  $D^j(UV) = D^j(U)D^j(V)$ ; for integer  $j$ , Theorem 11.5 makes  $D^\ell$  a function on SO(3) as well. In Euler coordinates, with  $U(\alpha, \beta, \gamma) = e^{-i\alpha J_z} e^{-i\beta J_y} e^{-i\gamma J_z}$ ,

$$D_{m'm}^j(\alpha, \beta, \gamma) = e^{-i(\alpha m' + \gamma m)} d_{m'm}^j(\beta), \quad d^j(\beta) = \text{matrix of } e^{-i\beta J_y}, \quad (11.6)$$

so the whole  $j$ -dependence sits in the single real matrix  $d^j$ . For instance  $d^{1/2}(\beta) = \begin{pmatrix} \cos \frac{\beta}{2} & -\sin \frac{\beta}{2} \\ \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix}$ , half-angles included.

One identity ties the matrices to Section 11.4. In the basis  $\{Y_{\ell m}\}$  of  $\mathcal{Y}_\ell$  the function action has matrix elements  $D^\ell: \Pi(R) Y_{\ell m} = \sum_{m'} D_{m'm}^\ell(R) Y_{\ell m'}$ . Only the  $m' = 0$  harmonic survives at the north pole — from  $\Pi(R_z(\theta)) Y_{\ell m'} = e^{-i\theta m'} Y_{\ell m'}$  evaluated at the fixed point  $\hat{z}$  — so evaluating the action there gives

$$D_{0m}^\ell(R) = c_\ell^{-1} Y_{\ell m}(R^{-1}\hat{z}), \quad c_\ell = Y_{\ell 0}(\hat{z}) = \sqrt{\frac{2\ell+1}{4\pi}} \text{ (standard normalisation)} : \quad (11.7)$$

one row of every integer Wigner matrix is a spherical harmonic in disguise.

**Example 11.17 (The spin-one Wigner matrix).** For  $j = 1$  the exponential series collapses exactly as it did for the Pauli matrices in Lecture 8: the spin-one  $J_y$  of Lecture 10 has eigenvalues  $0, \pm 1$ , so  $J_y^3 = J_y$  and the series folds into  $e^{-i\beta J_y} = \mathbb{K} - i \sin \beta J_y + (\cos \beta - 1) J_y^2$ , giving

$$d^1(\beta) = \begin{pmatrix} \frac{1+\cos \beta}{2} & -\frac{\sin \beta}{\sqrt{2}} & \frac{1-\cos \beta}{2} \\ \frac{\sin \beta}{\sqrt{2}} & \cos \beta & -\frac{\sin \beta}{\sqrt{2}} \\ \frac{1-\cos \beta}{2} & \frac{\sin \beta}{\sqrt{2}} & \frac{1+\cos \beta}{2} \end{pmatrix},$$

rows and columns ordered  $m = 1, 0, -1$ . Three readings. First, this is no stranger: it is the ordinary rotation  $R_y(\beta)$  rewritten in the harmonic basis  $\{-(x+iy)/\sqrt{2}, z, (x-iy)/\sqrt{2}\}$  of  $H_1$  — spin one is the vector representation (Sheet 10) in spherical dress. One column suffices to check:  $\Pi(R_y(\beta)) z = x \sin \beta + z \cos \beta$ , whose coordinates in that basis are exactly the middle column. Second, the middle row is the identity (11.7) at  $\ell = 1$ : for instance  $D_{00}^1 = \cos \beta = c_1^{-1} Y_{10}(R^{-1}\hat{z})$ , the polar angle of  $R^{-1}\hat{z}$  being  $\beta$ ; Sheet 11 checks the remaining entries, Euler phases included. Third, the endpoints:  $d^1(\pi)$  is the antidiagonal flip  $m \mapsto -m$  with signs  $(-1)^{1-m}$ , and

$$d^1(2\pi) = \mathbb{K}, \quad \text{while} \quad d^{1/2}(2\pi) = -\mathbb{K} :$$

Lecture 10's  $(-1)^{2j}$ , now wearing Euler coordinates.

Everything else about these matrices is computational and lives in Notebook 11 (WignerD) and on Sheet 11; Lecture 12 will need only their traces.

## 11.6 Rays, projective representations, and Wigner's theorem

One debt remains, and it is conceptual. Lecture 10 took the symmetry group of a quantum system to be SU(2) while geometry insisted on SO(3), and promised this lecture would arbitrate. The arbitration turns on what a quantum state actually is.



A state is not a vector. Two unit vectors differing by a phase,  $\psi$  and  $e^{i\alpha}\psi$ , are indistinguishable by any measurement: every prediction of quantum mechanics is built from transition probabilities

$$P([\varphi], [\psi]) = |\langle \varphi | \psi \rangle|^2,$$

which depend only on the rays  $[\psi] = \{e^{i\alpha}\psi : \alpha \in \mathbb{R}\}$ . The honest state space is the set of unit rays, and the honest notion of symmetry follows: a *quantum symmetry* is a bijection  $T$  of the rays preserving all transition probabilities. What such maps can be is settled by a theorem we import — the last conceptual import of the course.

**Theorem 11.18 (Wigner's theorem; imported).** *Every quantum symmetry  $T$  is implemented by an operator  $U: \mathcal{H} \rightarrow \mathcal{H}$  with  $T[\psi] = [U\psi]$ , where  $U$  is either unitary or antiunitary — additive, conjugate-linear, with  $\langle U\varphi | U\psi \rangle = \overline{\langle \varphi | \psi \rangle}$  — and  $U$  is unique up to a phase.*

*Reference for the omitted proof:* Weinberg, *The Quantum Theory of Fields I*, Appendix 2.A.

Antiunitary implementers are not exotica — time reversal is one, as Sheet 7 noted — but they cannot occur in a connected symmetry group: every element of a connected matrix Lie group is a finite product of exponentials (Lecture 8), hence of squares  $e^X = (e^{X/2})^2$ , and the implementer of a square,  $U_{h^2} = \omega U_h U_h$ , is unitary whether  $U_h$  is unitary or antiunitary. For rotations, then, only unitaries — but unitaries defined up to phases, and the phases are the whole story.

**Definition 11.19 (Projective representation).** *A projective unitary representation of a group  $G$  on  $\mathcal{H}$  is a map  $g \mapsto U_g$  into the unitaries with*

$$U_g U_h = \omega(g, h) U_{gh}, \quad |\omega(g, h)| = 1 : \quad (11.8)$$

a representation up to phases — exactly what Wigner's theorem delivers when a group acts by quantum symmetries.

**Proposition 11.20 (Half-integer spin is projectively lawful).** *For every  $j$ , choose for each  $R \in SO(3)$  a preimage  $U_R \in \Phi^{-1}(R)$  and set  $W_R = \Pi_j(U_R)$ . Then*

$$W_R W_{R'} = \pm W_{RR'} :$$

*a projective unitary representation of  $SO(3)$ , whose action on rays is independent of all choices. For integer  $j$  the signs can be removed (Theorem 11.5); for half-integer  $j$  they cannot: no honest representation of  $SO(3)$  induces the same ray action.*

*Proof.* Since  $U_R U_{R'} \in \Phi^{-1}(RR') = \{\pm U_{RR'}\}$  and  $\Pi_j(-U) = (-1)^{2j} \Pi_j(U)$  (Lecture 10), the multiplication rule holds with signs, and replacing  $U_R$  by  $-U_R$  changes  $W_R$  by at most a sign — invisible on rays. Suppose now an honest representation  $\tilde{\Pi}$  of  $SO(3)$  induced the same ray action. For each  $U \in SU(2)$  the unitaries  $\tilde{\Pi}(\Phi(U))$  and  $\Pi_j(U)$  implement the same ray map, so by the uniqueness in Wigner's theorem  $\tilde{\Pi}(\Phi(U)) = \lambda(U) \Pi_j(U)$  for a phase  $\lambda(U)$ ; being a ratio of homomorphisms,  $\lambda$  is a continuous homomorphism  $SU(2) \rightarrow U(1)$ . Its derivative is a Lie algebra map  $\mathfrak{su}(2) \rightarrow \mathfrak{u}(1)$  into an abelian algebra, and it must kill all brackets — but brackets span  $\mathfrak{su}(2)$  ( $X_1 = [X_2, X_3]$  and cyclically), so the derivative vanishes and  $\lambda \equiv 1$  on the connected  $SU(2)$ . Then  $\tilde{\Pi} \circ \Phi = \Pi_j$ , forcing  $\Pi_j(-\kappa) = \tilde{\Pi}(\kappa) = \kappa$  — false for half-integer  $j$ . ■

Here, at last, is the arbitration. Quantum mechanics is formulated on rays; its symmetry groups act by projective representations; and the projective representations of  $SO(3)$  are

precisely the honest representations of  $SU(2)$  — the half we proved above, the converse (every projective representation of  $SO(3)$  arises so, after adjusting phases) being Bargmann’s theorem, stated as Theorem 11.21 below and imported. *The physicist’s rotation group* is  $SU(2)$ , not by decree but because the quantum kinematics of rays makes the simply connected cover available free of charge. Lecture 10’s IOU is paid in full: a spin- $\frac{1}{2}$  system violates nothing; it merely uses the phases that vectors carry and rays forgive. (Not every group is so lucky: for the Galilei group genuinely irremovable phases appear, carrying the particle’s mass — close kin of the central element of Lecture 13’s Heisenberg algebra. For the Lorentz group, Lecture 14 will find the same luck as here.)

**Theorem 11.21 (Bargmann; imported).** *Let  $G$  be a connected Lie group whose Lie algebra  $\mathfrak{g}$  has vanishing second cohomology,  $H^2(\mathfrak{g}, \mathbb{R}) = 0$ . Then every continuous projective unitary representation of  $G$  — every assignment  $g \mapsto W_g$  obeying  $W_g W_{g'} = c(g, g') W_{gg'}$  with phases  $c(g, g') \in U(1)$  — is, on rays, induced by an honest strongly continuous unitary representation of the universal cover  $\tilde{G}$ : the phases  $c(g, g')$  can be removed after lifting to  $\tilde{G}$ . The hypothesis holds automatically when  $\mathfrak{g}$  is semisimple — in particular for  $\mathfrak{g} = \mathfrak{su}(2)$ , by Whitehead’s lemma  $H^2(\mathfrak{g}, \mathbb{R}) = 0$ .*

*Reference for the omitted proof:* Bargmann, *On unitary ray representations of continuous groups*, Ann. of Math. **59** (1954) 1–46; Weinberg, *The Quantum Theory of Fields I*, §2.7.

Applied to  $SO(3)$ , whose universal cover is  $SU(2)$  with  $\mathfrak{su}(2)$  semisimple, this supplies exactly the converse used above: every projective representation of  $SO(3)$  arises, after adjusting phases, from an honest representation of  $SU(2)$ . (The Galilei group fails the hypothesis —  $H^2 \neq 0$ , the surviving cocycle being the mass — which is why its phases do not lift. For the connected Lorentz group the same argument uses the semisimple Lorentz algebra and lifts to its  $SL(2, \mathbb{C})$  cover, as Lecture 14 explains. The full Poincaré algebra is not semisimple; its distinct cohomological lifting input is stated at the point of use in Lecture 15.)

And the minus sign is not philosophy. In 1975 two independent crystal-neutron-interferometer experiments made it relative and therefore observable. Rauch and collaborators coherently split unpolarised slow neutrons, varied the unequal magnetic-field integrals along the two paths, and observed output-intensity oscillations with the predicted  $4\pi$  spinor period. Werner and collaborators instead used a polarised beam and measured the interferometric phase shift produced by magnetic precession, directly identifying the sign reversal after a  $2\pi$  rotation. Both experiments use magnetic precession and path recombination, but they are not the same protocol. The  $(-1)^{2j}$  of Lecture 10, computed from bare algebra, is visible in a reactor hall.

*Primary experimental sources:* H. Rauch et al., “Verification of coherent spinor rotation of fermions,” *Physics Letters A* **54** (1975), 425–427, doi:10.1016/0375-9601(75)90798-7; S. A. Werner et al., “Observation of the phase shift of a neutron due to precession in a magnetic field,” *Physical Review Letters* **35** (1975), 1053–1055, doi:10.1103/PhysRevLett.35.1053.

The season’s geography is complete: one algebra, two groups, a two-to-one bridge between them, and quantum mechanics choosing the upper deck. Lecture 12 restarts the engines of Part I — invariant integration in place of group sums, characters (including the crystal-field formula Part I borrowed on credit), and the question the experimentalists ask daily: what happens when two angular momenta meet. Lecture 14 will replay today’s construction one signature up, with  $SL(2, \mathbb{C})$  standing over the Lorentz group.

**Remark 11.22 (Computer algebra).** Notebook 11 checks SphericalHarmonicY and WignerD

against our ladder constructions; verifies  $\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1)Y_{\ell m}$  numerically; plots  $|Y_{\ell m}|$  over the sphere; and animates the double cover — a path of  $SU(2)$  matrices beside its image rotations, the  $2\pi$  loop arriving at  $-\mathbb{I}$  while the  $4\pi$  loop closes — with the belt trick rendered on screen.

### Summary.

- *Double cover.*  $\Phi : SU(2) \rightarrow SO(3)$  is a two-to-one homomorphism with kernel  $\{\pm \mathbb{I}\}$  (Theorem 11.1);  $SO(3)$  has fundamental group  $\mathbb{Z}/2$  (Proposition 11.3).
- *Descent.* The representations of  $SO(3)$  are exactly the integer-spin representations of  $SU(2)$  (Theorem 11.5).
- *Spherical harmonics.* The degree- $\ell$  harmonic polynomials carry the spin- $\ell$  representation, of dimension  $2\ell+1$  (Theorem 11.9, Proposition 11.8), with spherical-Laplacian eigenvalue  $-\ell(\ell + 1)$  (Theorem 11.12).
- *Completeness.* The  $Y_{\ell m}$  form an orthonormal basis of  $L^2(S^2)$  (Theorem 11.14); Euler angles and the Wigner matrices coordinatise rotations (Proposition 11.15).
- *Projective representations.* Quantum states are rays (Wigner's theorem, Theorem 11.18), so half-integer spin is lawful and  $SU(2)$  is the operative group (Proposition 11.20).

**Tutorial.** Exercise Sheet 11.

### References

Hall §4.7 ( $SO(3)$  versus  $SU(2)$ ); Stillwell Ch. 2; Tung (the  $SO(3)$  and  $SU(2)$ – $SO(3)$  chapters); Elliott–Dawber (rotation-group sections); Woit (rotations and spin); Costa–Fogli Ch. 2. For Bargmann's theorem (Theorem 11.21): Bargmann, *Ann. of Math.* **59** (1954) 1–46; Weinberg, *QFT I* §2.7. For the 1975 neutron experiments: Rauch et al., *Phys. Lett. A* **54**, 425–427; Werner et al., *Phys. Rev. Lett.* **35**, 1053–1055.

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